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GROUP: Technical

SECTION: Section A

MARKS: 1

Instructions

Answer Correct

TIME LEFT: 00:12:52

Consider an unsorted list of 8 elements as given below:

23 61 73 82 9 43 11 3

What will be the sequence after two passes of quick sort if first element of the list is chosen as pivot always and the list is sorted in ascending order?

[Note: Both left and right recursive calls are made together]

☐

3 9 11 23 43 61 82 73

☒

61 73 82 23 9 43 11 3

☐

9 3 11 23 82 43 73 61

☐

3 9 11 23 82 43 73 61

Previous Question

Next Question

QUESTION: 30

GROUP: Technical

SECTION: Section C

Marked: 1

Instructions

Consider the given segment map table along with the logical addresses for the corresponding segments

Segment	Base	Length
0	90	600
1	1992	14
2	569	256
3	2669	123
4	912	569

Consider the following logical address (0, 430), (1, 236), (2, 200), (4, 100) Which request is invalid?

☐ (2, 200)

☐ (4, 100)

☐ (1, 236)

☐ (0, 430)

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QUESTION: 31

GROUP Technical

SECTION Section A

Mark(s) 1

Instructions

Consider the following pseudo-code algorithm segment.
algo-1(i) {
 for all vertices u {
 num(u) = 0
 edges = null
 }
 i = 1
 while there is a vertex v such that num(v) == 0 {
 num(v) = i++
 enqueue(v)
 while queue is not empty {
 v = dequeue()
 for all vertices u adjacent to v {
 if num(u) == 0 {
 num(u) = i++
 enqueue(u)
 attach edge (uv) to edges
 }
 }
 }
 }
}

What does the above algorithm describe?

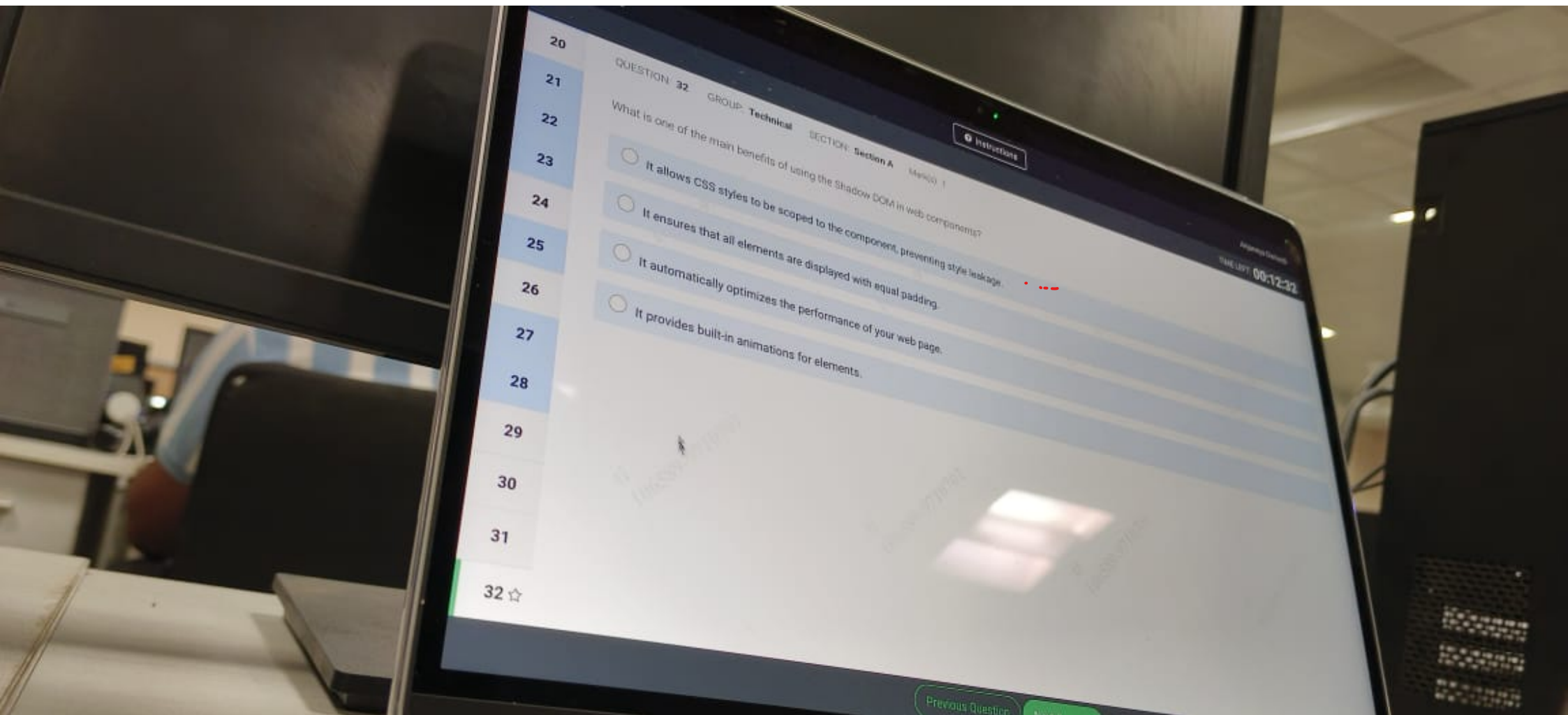
- ☐ Array based implementation of queue
- ☐ Depth first traversal

BFS

Previous Question

Next Question

TIME LEFT 00:12:37



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QUESTION 32

GROUP: Technical

SECTION: Section A

Marked 1

Instructions

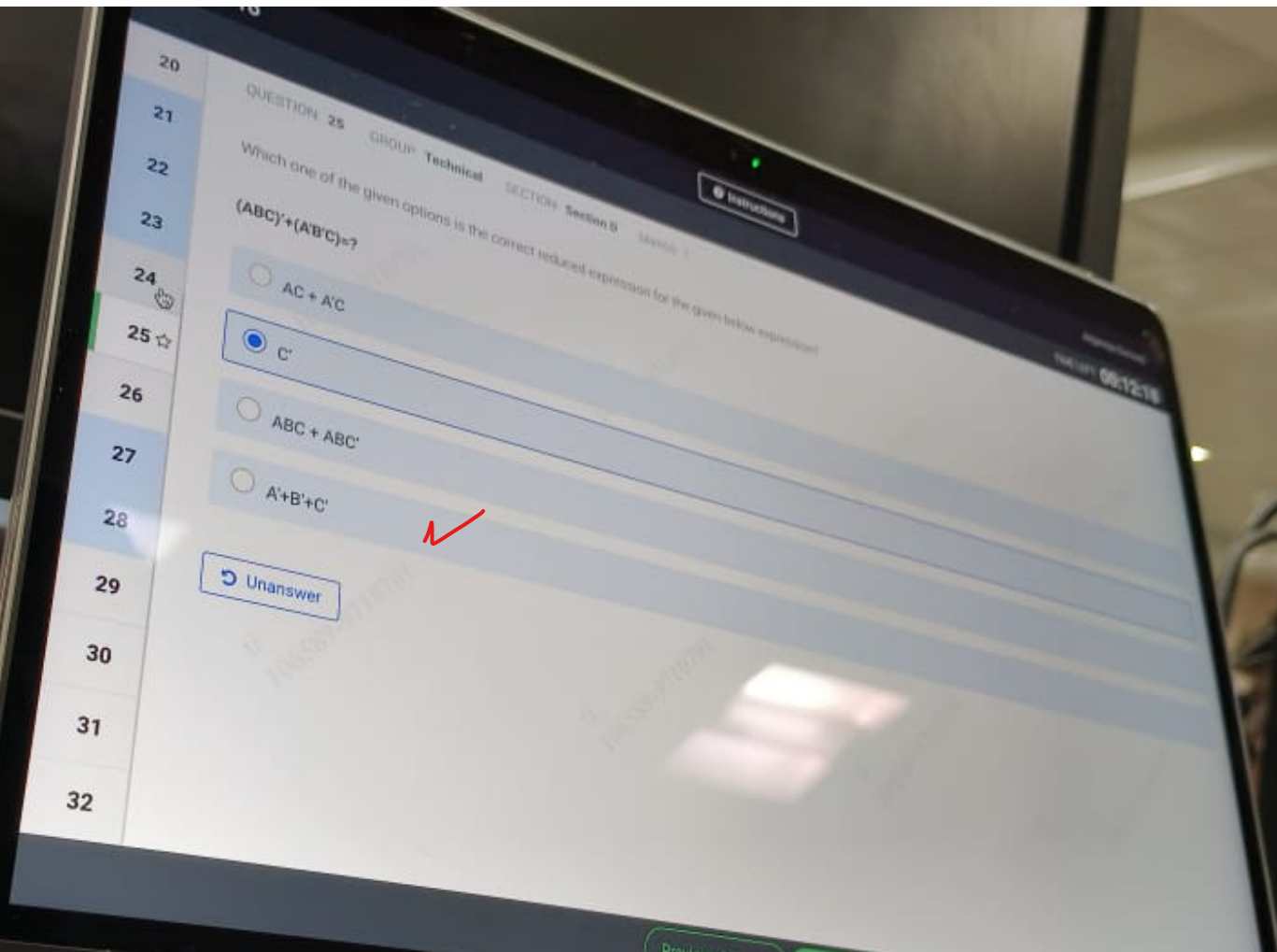
Time Left 00:12:32

What is one of the main benefits of using the Shadow DOM in web components?

- ☒ It allows CSS styles to be scoped to the component, preventing style leakage.
- ☐ It ensures that all elements are displayed with equal padding.
- ☐ It automatically optimizes the performance of your web page.
- ☐ It provides built-in animations for elements.

Previous Question

Next Question



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QUESTION 29

GROUP: Technical

SECTION: Section A

Attempts: 1

Instructions

Consider an unsorted list of 8 elements as given below:

23 61 73 82 9 43 11 3

What will be the sequence after two passes of quick sort if first element of the list is chosen as pivot always and the list is sorted in ascending order?

[Note: Both left and right recursive calls are made together]

☐ 3 9 11 23 43 61 82 73

☐ 61 73 82 23 9 43 11 3

☐ 9 3 11 23 82 43 73 61

☐ 3 9 11 23 82 43 73 61

Previous Question

Next Question

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GROUP: Aptitude

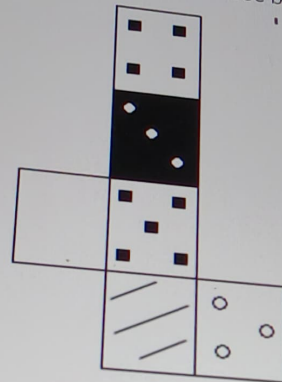
SECTION: Puzzles

Mark(s): 1

Attempts: 0

TIME LEFT: 00:02:21

The following figure marked (X) has been folded to produce different box patterns. Also, four boxes are given below marked (1), (2), (3) and (4). Choose the correct alternative which indicates that those boxes are produced by folding the figure (X).



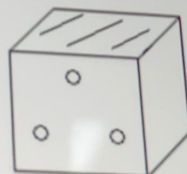
(X)



(1)



(2)



(3)



(4)

☐ (2), (3) and (4)

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void Insert(struct Node *Start)
{
 struct node *ptr, *temp;
 ptr = CreateNode(); // CreateNode() creates a node.
 if(Start == NULL)
 Start = ptr;
 else
 {
 temp = Start;
 Start = ptr;
 ptr->link = temp;
 }
}

- ☐ No element is inserted.
- ☐ Creates a Wild pointer and hence Run Time Error.
- ☐ Insert an element at the last position.
- ☒ Insert an element at the first position. ✓

Instructions

Average Grade
TIME LEFT 00:15:51

Previous Question

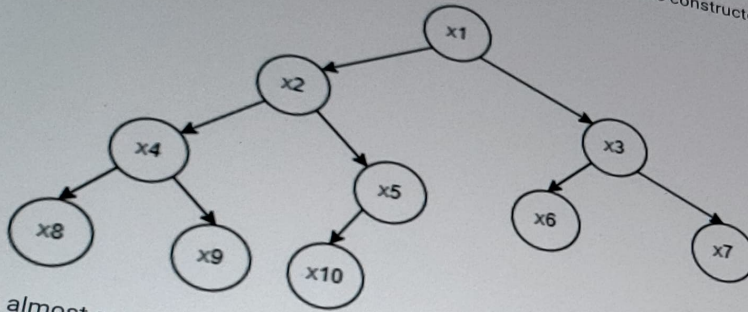
Next Question

33

34 ☆

Consider that you are provided with a list of A of N integers. Construct the almost complete binary tree by using them and write a program to print the sum S of all the elements present in the given level.

If the elements of the list L are $[x_1, x_2, x_3, x_4, x_5, x_6, \dots, x_{10}]$ then the constructed almost binary tree would be in the following manner.



An almost complete binary tree is a binary tree that satisfies the following conditions:

- Insertion of nodes must take place level by level and all the nodes must be left justified.
- All the levels from 1 to $h-1$ level (h stands for the number of levels) should be completely filled without any gap.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output.

Constraints:

Element should not be negative

Input Format:

The first line of input contains N where N denotes the total number of elements in the list A

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33 ☆

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"Meeting people frequently is the surest way to get re-elected", advised the party President. Margaret, one of the party's elected representatives, decided to take the advice seriously and went on a tour of her constituency. One day, she came across a particular street, where there were N houses. Instead of visiting every house, which would take a lot of time, she decided to randomly visit at least V out of each C consecutive houses on the street.

Before starting off, she is wondering if there are multiple ways to select the houses for this 'meet & greet'. Write a program to help Margaret know how many different ways she can visit the houses.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings anywhere in the program, as these contribute to the standard output and test cases will fail.

Constraints: $1 \leq N \leq 10^9$
 $1 \leq V \leq C \leq N$ **Input Format:**

A single line of input contains three space-separated integers N , C and V .

Output Format:

Output should contain a single integer denoting the possible number of ways to visit the houses. As the number can get exponentially huge, return its modulo 10^9+7 .

Sample Input 1:

4 2 1

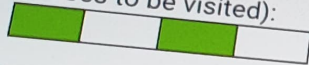
Sample Output 1:

3

Explanation:

$N = 4$, $C = 2$ and $V = 1$.

From a total 4 houses, Margaret wants to visit at least 1 house from each 2 consecutive houses. There are 3 different ways to do this, as follows (green houses are the houses to be visited):

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